MIL-M-7404D
4 October 1968
SUPERSEDING
MIL-M-7404C(USAF)
28 July 1966

# MILITARY SPECIFICATION

## MAINTENANCE PLATFORM, AIRCRAFT, ADJUSTABLE, 3 TO 7 FEET, TYPE B-4A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of adjustable aircraft maintenance platform, designated type B-4A.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| QQ-Z-325 | Zinc Coating, Electrodeposited, Requirements For |
| UU-T-81 | Tags, Shipping And Stock |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-636 | Box, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-D-1000 | Drawings, Engineering And Associated Lists |
| MIL-C-5501 | Caps And Plugs, Protective, Dust And Moisture Seal |
| MIL-H-5606 | Hydraulic Fluid, Petroleum Base; Aircraft, Missile, And Ordnance |
| MIL-H-6083 | Hydraulic Fluid, Petroleum Base, For Preservation And Testing |
| MIL-W-8611 | Welding, Metal Arc And Gas, Steels, And Corrosion And Heat Resistant Alloys; Process For |



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STANDARDS

Military

| | |
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| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of US Military Property |
| MIL-STD-143 | Specifications And Standards Order Of Precedence For The Selection Of |
| MIL-STD-162 | Materials Handling Equipment: Preparation For Shipment, Storage, Cyclic Maintenance, Routine Testing And Processing |
| MIL-STD-831 | Test Reports, Preparation Of |
| MIL-STD-1186 | Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods |
| MS33540 | Safety Wiring And Cotter Pinning, General Practice For |

DRAWING

Air Force

| | |
|---|---|
| 54J6345 | Maintenance Platform Assembly, Aircraft, Adjustable, 3 to 7 Feet, Type B-4A |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 AN or MIL standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they develop suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and if practicable, on the contractor's drawings. In case there is no suitable corresponding AN or MIL standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.



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3.3 Design and construction. Design and construction of the platform shall conform to Drawing 54J6345. When the drawing and this specification conflict, this specification shall govern.

3.3.1 Welding. All welding shall be performed in accordance with MIL-W-8611.

3.3.2 Threaded parts. Threaded parts shall be positively locked or safety wired in accordance with MS33540.

3.3.3 Hydraulic system. The hydraulic system shall be equipped with a reservoir and filler opening located on the top. A drain plug shall be provided. An air vent shall be incorporated in the filler cap. Means shall be provided in the reservoir tank to prevent the oil from splashing out when the maintenance platform is moved. An oil strainer shall be installed in the filler opening. The hydraulic system and reservoir shall be filled with hydraulic fluid conforming to MIL-H-6083 for shipment and storage.

3.3.3.1 Actuating pump. The actuating pump used in height adjustment of the platform shall be designed for ease of maintenance, operation, and for resistance to the temperatures specified herein. The force required at the end of the handle of the pump to raise the platform from the low level shall not exceed 15 pounds with 150 pounds on the platform. A maximum of a hundred strokes shall be required to accomplish the first 4 feet of adjustment.

3.3.3.2 Hydraulic cylinder. Wherever practicable, the hydraulic components shall conform to AN or MIL standards. A seal shall be installed to prevent entrance of moisture into the cylinder. All hydraulic fittings and lines (rubber or metal) shall be suitable for operation in temperatures ranging from -65° Fahrenheit (F) to +160°F. Low temperature synthetic rubber packing shall be used. The plunger or piston shall be chromium plated.

3.4 Performance.

3.4.1 Static load. The maintenance platform shall be capable of withstanding a uniformly distributed load of 1,000 pounds with no permanent deformation or deflection of any member of the platform. The expanded metal deck shall be capable of sustaining a 300 pound load distributed over an area of 25 square inches without deflection or depression of more than $\frac{1}{4}$ inch.

3.4.1.1 Steps. The steps or rungs shall be capable of withstanding a load of 400 pounds without damage or permanent deformation of the steps.

* 3.4.2 Wheel locks. With all wheel locks fully engaged, no damage to the locks or slipping of wheels shall result when the maintenance platform is moved a distance of 5 feet on smooth, dry concrete.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.



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3.6 Finish.

3.6.1 Plating. All bolts, nuts, and pins used in the construction of the platform shall be cadmium plated in accordance with QQ-P-416 or zinc plated in accordance with QQ-Z-325.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship.

3.8.1 General. The maintenance platform, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of welding, brazing, painting, riveting, and freedom of parts from burrs and sharp edges.

3.8.1.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.8.2 Cleaning. The maintenance platform shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material during and after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the maintenance platform shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests.

4.3 Test conditions. The hydraulic system shall be filled with hydraulic fluid conforming to MIL-H-5606 for the tests.

4.4 Preproduction testing.

4.4.1 Test sample. The preproduction test sample shall consist of one platform representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract or



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purchase order, at the contractor's plant under the supervision of the procuring activity (see 6.2).

4.4.2 Test report. Upon completion of preproduction tests, the contractor shall submit a test report prepared in accordance with MIL-STD-831.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests in 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling tests.

4.5.1 Individual tests. Individual tests shall consist of examination of product in 4.6.1.

* 4.5.2 Sampling tests. One maintenance platform shall be selected at random from every fifty or fraction thereof produced and subjected to the following tests:

a. Static load test (see 4.6.2).

b. Ten percent (ten cycles) of the endurance test (see 4.6.3).

4.5.2.1 Rejection and retest. When one or more items from a production run fail to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all the sampling tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or items later produced shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a sampling test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The maintenance platform shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, marking, and finish.



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* 4.6.2 Static load test. The maintenance platform shall be tested at all heights throughout its range in increments not to exceed 6 inches. At each of these increments, a uniformly distributed load of 1,000 pounds shall be placed on the platform. No member of the maintenance platform shall be depressed or deflected more than 1½ inches under this load, nor shall any permanent deformation or deflection of any member of the maintenance platform be evident after completion of this test. Safety pins shall not be installed for this test.

* 4.6.2.1 The expanded metal deck shall be subjected to a concentrated vertical load of 300 pounds distributed over an area of 25 square inches. No point on the expanded metal deck shall be depressed or deflected more than ¼ inch when subjected to this test.

* 4.6.2.2 Each step or rung shall be individually tested by applying a load of 400 pounds at or near the center of the step. No evidence of damage or permanent deformation of the steps shall be apparent at the conclusion of this test.

* 4.6.2.3 Upon completion of the static load test, the maintenance platform shall be visually inspected. There shall be no leakage of hydraulic fluid, elongation of bolt holes, or failure of welding and materials.

* 4.6.3 Endurance. The maintenance platform shall operate satisfactorily after it has been subjected to a hundred cycles of raising a 500-pound load to maximum height and lowering the load to minimum height.

* 4.6.4 Settling test. With the platform raised 20 percent above the minimum height and with a platform load of 1,000 pounds, the rate of settling of the hydraulic ram shall not exceed 0.020 inch per hour. Safety pins shall not be installed for this test.

* 4.6.5 Wheel locks. With all wheel locks fully engaged, the maintenance platform shall be subjected to a drawbar pull sufficient to move the unloaded maintenance platform a distance of 5 feet on smooth, dry concrete. There shall be no indication of retraction, deformation, or damage to the locks or rolling of wheels.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage, loss or breakage in transit and to accomplish reduction in cube. Bolts, nuts and pins and other attaching parts removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and



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mating parts shall be matchmarked. Matchmarking shall be printed on shipping tags conforming to UU-T-81, type A. The printed tags shall be waterproofed in accordance with MIL-STD-129.

5.1.2 Level A. The maintenance platform shall be preserved in accordance with MIL-P-116, method I. All bare or unpainted metal surfaces subject to rust or corrosion shall be coated with type P1 preservative in accordance with MIL-P-116. The hydraulic systems shall be preserved in accordance with MIL-STD-162. All openings shall be capped or plugged with MIL-C-5501 closures.

5.1.3 Level C. Preservation and packaging at this level shall afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets requirements of this level.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. The maintenance platform preserved and packaged as specified in 5.1 shall be packed in a container conforming to PPP-B-601, style A, overseas type, and surface-treated in accordance with the requirements of the specification. Containers shall be provided with three nominal 4 inch by 4 inch skids which extend the full exterior width of the box and meet the requirements for PPP-B-601. All loose parts shall be packed in a domestic type container conforming to PPP-B-636 and secured within the PPP-B-601 container. Containers shall be closed and strapped in accordance with the appendix of applicable specifications.

5.2.2 Level B. Same requirements as Level A, except the containers shall be domestic type.

5.2.3 Level C. The maintenance platform packaged as specified in 5.1 shall be packed in a manner which affords adequate protection at the lowest rate against damage during direct shipment from the supply to the first receiving activity for immediate use. The level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Physical protection. All cushioning, blocking, bracing and bolting shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The type B-4A maintenance platform covered by this specification is intended for use in the field for maintenance of aircraft under tactical conditions.



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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Required levels of preservation, packaging, and packing (see 5.1 and 5.2).

c. Location of preproduction testing (see 4.4.1).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 1730-0103



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